

ANSWERS (BOTH PARTS)

- (1) Find the eigenvalues and eigenvectors of the matrix $A = \begin{bmatrix} 3 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 3 \end{bmatrix}$. Find an orthogonal matrix S and a diagonal matrix D such that $S^{-1}AS = D$.

Answer:

The characteristic polynomial of A is $p_A(\lambda) = \det \begin{bmatrix} 3-\lambda & -1 & 0 \\ -1 & 2-\lambda & -1 \\ 0 & -1 & 3-\lambda \end{bmatrix} = (3-\lambda)(\lambda^2 - 5\lambda + 5) + (-1)(3-\lambda) = (3-\lambda)(4-\lambda)(1-\lambda)$.

Hence the eigenvalues of A are 1, 3 and 4 and the corresponding unit eigenvectors are $(\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}})$, $(\frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}})$ and $(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}})$.

Hence $S^{-1} = S^t = \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{bmatrix}$ and D is the diagonal matrix $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix}$.

- (2) Let $f(x, y) = (\cos(x), e^y, x^2 + y^2)$ and $g(x, y, z) = (\sin(z), x + y^4)$. Using the chain rule for derivatives of vector fields, find $D(g \circ f)(0, 0)$.

Answer: $D(g \circ f)(0, 0) = Dg(f(0, 0)) \circ Df(0, 0) = Dg(1, 1, 0) \circ Df(0, 0)$ by Chain Rule (since $f(0, 0) = (1, 1, 0)$).

$Dg(x, y, z) = \begin{bmatrix} 0 & 0 & \cos(z) \\ 1 & 4y^3 & 0 \end{bmatrix}$. Hence, $Dg(1, 1, 0) = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 4 & 0 \end{bmatrix}$.

$Df(x, y) = \begin{bmatrix} -\sin(x) & 0 \\ 0 & e^y \\ 2x & 2y \end{bmatrix}$. Hence, $Df(0, 0) = \begin{bmatrix} 0 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$.

Thus, $D(g \circ f)(0, 0) = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 4 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 4 \end{bmatrix}$.

- (3) Compute the double integral $\int \int_S \sqrt{1 - x^2 - y^2} dx dy$, where $S = \{(x, y) | x^2 + y^2 \leq 1, x \geq 0, y \geq 0\}$ in two ways, directly and using polar coordinates (and the change of variable theorem for double integrals).

Answer: (Direct method) Let $a = \sqrt{1 - y^2}$, hence $a^2 = 1 - y^2$.

Then $\int \int_S \sqrt{1-x^2-y^2} dx dy = \int_0^1 \int_0^{\sqrt{1-y^2}} \sqrt{1-x^2-y^2} dx dy = \int_0^1 \int_0^a \sqrt{a^2-x^2} dx dy = \int_0^1 \int_0^a a \sqrt{1-\frac{x^2}{a^2}} dx dy$. Let $\frac{x}{a} = \sin(t)$, then $dx = a \cos(t)$ and $\sqrt{1-\frac{x^2}{a^2}} = \cos(t)$. Also as x varies from 0 to a , t varies from 0 to $\pi/2$. Hence the inner integral $\int_0^a a \sqrt{1-\frac{x^2}{a^2}} dx = \int_0^{\pi/2} a^2 \cos^2(t) dt = \frac{a^2}{2} \int_0^{\pi/2} (\cos(2t) + 1) dt = \frac{\pi a^2}{4} = \frac{\pi(1-y^2)}{4}$. Thus the double integral is $\frac{\pi}{4} \int_0^1 (1-y^2) dy = \frac{\pi}{6}$.

(Polar coordinates) Using the change of variable theorem, $\int \int_S \sqrt{1-x^2-y^2} dx dy = \int_0^1 \int_0^{\pi/2} r \sqrt{1-r^2} d\theta dr = \frac{\pi}{2} \int_0^1 r \sqrt{1-r^2} dr = \frac{\pi}{2} [-\frac{1}{3}(1-r^2)^{\frac{3}{2}}]_{r=0}^1 = \frac{\pi}{6}$.

- (4) Consider the map $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ given by $f(x, y) = x^2 - xy + y^2 - 2x + y$. Find the directional derivative of f at $(1, 1)$ along the direction $(2, 3)$. Find all the stationary points of f . Classify these stationary points as relative maxima or relative minima or saddle point.

Answer: The gradient of f at $(1, 1)$ is $(\frac{\partial f}{\partial x}(1, 1), \frac{\partial f}{\partial y}(1, 1)) = (2x - y - 2, -x + 2y + 1)_{((x,y)=(1,1))} = (-1, 2)$. Hence the directional derivative along the direction $(2, 3)$ is the dot product of $(-1, 2)$ and the unit vector in the direction of $(2, 3)$ which is $\frac{(2,3)}{\sqrt{13}}$. Hence the answer is $\frac{(-1,2) \cdot (2,3)}{\sqrt{13}} = \frac{4}{\sqrt{13}}$.

To solve for stationary points, we solve for: $\frac{\partial f}{\partial x} = 2x - y - 2 = 0$ and $\frac{\partial f}{\partial y} = -x + 2y + 1 = 0$. Hence there is only one stationary point of f , namely $(x, y) = (1, 0)$.

To classify the stationary point, we consider the second derivatives of f at $(1, 0)$. We have $\frac{\partial^2 f}{\partial x^2} = 2$, $\frac{\partial^2 f}{\partial y^2} = 2$ and $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x} = -1$. Hence the Hessian matrix is $\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$. Now the determinant and the top left entry are both positive, hence $(1, 0)$ is a relative minimum of f .

- (5) Find the radius of convergence for the power series $\sum_{n=1}^{\infty} (\frac{3^n}{n4^n}) x^n$.

Answer: $\beta = \limsup (\frac{3^n}{n4^n})^{\frac{1}{n}} = \frac{3}{4}$, hence $R = \frac{1}{\beta} = \frac{4}{3}$.

- (6) For $x \in [0, \infty)$, let $f_n(x) = \frac{x}{n}$, find the pointwise limit function f of the sequence (f_n) and determine whether $f_n \rightarrow f$ uniformly on $[0, \infty)$.

Answer: For any $x \in [0, \infty)$, $\lim_{n \rightarrow \infty} f_n(x) = 0$. Hence the pointwise limit function of the sequence (f_n) is the zero function $f(x) = 0$ for all $x \in [0, \infty)$.

The sequence f_n does not converge to the zero function f uniformly on $[0, \infty)$. Because if the convergence was uniform, then given any $\epsilon > 0$, there would exist N such that for all $n \geq N$, we must have $|f_n(x) - f(x)| = \frac{x}{n} < \epsilon$ for all $x \in [0, \infty)$. But that implies $x < N\epsilon$ for all $x \in [0, \infty)$ which is absurd.

- (7) Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be $f(x, y) = \frac{xy(x^2-y^2)}{x^2+y^2}$ if $(x, y) \neq (0, 0)$ and $f(0, 0) = 0$. Compute $\frac{\partial^2 f}{\partial x \partial y}(0, 0)$ and $\frac{\partial^2 f}{\partial y \partial x}(0, 0)$.

Answer: We will show that $\frac{\partial^2 f}{\partial x \partial y}(0, 0) = 1$ and $\frac{\partial^2 f}{\partial y \partial x}(0, 0) = -1$. We need to first compute $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$.

If $(x, y) \neq (0, 0)$, then $\frac{\partial f}{\partial x}(x, y) = \frac{(x^2+y^2)(3x^2y-y^3)-2x^2y(x^2-y^2)}{(x^2+y^2)^2}$.

Hence if $h \neq 0$, we have $\frac{\partial f}{\partial x}(0, h) = \frac{(0+h^2)(0-h^3)-0}{(0+h^2)^2} = \frac{-h^5}{h^4} = -h$.

Also $\frac{\partial f}{\partial x}(0, 0) = \lim_{h \rightarrow 0} \frac{f(0, h) - f(0, 0)}{h} = 0$.

Hence $\frac{\partial^2 f}{\partial y \partial x}(0, 0) = \lim_{h \rightarrow 0} \frac{\frac{\partial f}{\partial x}(0, h) - \frac{\partial f}{\partial x}(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{-h}{h} = -1$.

If $(x, y) \neq (0, 0)$, then $\frac{\partial f}{\partial y}(x, y) = \frac{(x^2+y^2)(x^3-3xy^2)-2xy^2(x^2-y^2)}{(x^2+y^2)^2}$.

Hence if $h \neq 0$, we have $\frac{\partial f}{\partial y}(h, 0) = \frac{(h^2+0)(h^3-0)-0}{(h^2+0)^2} = \frac{h^5}{h^4} = h$.

Also $\frac{\partial f}{\partial y}(0, 0) = \lim_{h \rightarrow 0} \frac{f(0, h) - f(0, 0)}{h} = 0$. Hence

Hence $\frac{\partial^2 f}{\partial x \partial y}(0, 0) = \lim_{h \rightarrow 0} \frac{\frac{\partial f}{\partial y}(h, 0) - \frac{\partial f}{\partial y}(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{h}{h} = 1$.